European Journal of Operational Research 272 (2019) 945–961

Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/ejor

Production, Manufacturing and Logistics

A stochastic multi-period industrial hazardous waste location-routing problem: Integrating NSGA-II and Monte Carlo simulation

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ARTICLE INFO

Article history:
Received 11 November 2017
Accepted 13 July 2018
Available online 19 July 2018

Keywords:
Combinatorial optimization
Industrial hazardous waste
Location-routing problem
Simulation optimization
Simheuristic

ABSTRACT

The present study extends a multi-objective mathematical model in the context of industrial hazardous waste management, which covers the integrated decisions of three levels with locating, vehicle routing, and inventory control. Analyzing these decisions simultaneously not only may lead to the most effective structure in the waste management network, but also may reduce the potential risk of managing the hazardous waste. Furthermore, because of the inherent complexity of the waste management system, uncertainty is inevitable and should be acknowledged to guarantee reliability in the decision-making process. From this perspective, the proposed model is novel in the following three aspects: (1) shifting from a deterministic to stochastic environment; (2) considering a multi-period planning horizon; and (3) incorporating the inventory decisions into the problem. The problem is formulated as a multi-objective stochastic Mixed-Integer Nonlinear Programming (MINLP) model, which can be easily converted into a MILP one. In terms of methodological contribution, a new simheuristic approach that is an integration of Non-Dominated Sorting Genetic Algorithm-II (NSGA-II) and Monte Carlo simulation is developed to overcome the stochastic combinatorial optimization problem of this study. Our findings verify the efficiency of the proposed approach as it is able to find a high-quality solution within a relatively reasonable computational time.

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1. Introduction

Any industrial waste which possesses properties like toxicity, corrosiveness, ignitability and chemical reactivity can be regarded as a hazardous waste. Industrial and manufacturing processes are the major sources of waste production. This is especially the case in countries such as Germany, where more than 75% of hazardous waste originates from the manufacturing industry (Williams, 2005). Management of these materials is a critical issue due to their imposed risk on the environment and human life as well. Collecting, transporting, treating, recycling and disposing of hazardous waste in a safe, efficient and cost-effective manner reflect the definition of hazardous waste management.

Beginning from the early work of Zografros and Samara (1989) on integrating the location and vehicle routing decisions in hazardous waste transportation and disposal, a rich literature has emerged on this topic. They formulated a multi-objective model to meet several objectives concerning travel time, transportation and disposal risk in a waste management system restricted to only one type of waste. List and Mirchandani (1991) proposed a model for multiple types of hazardous wastes. Their model considered risk as a function of external impacts, which is inversely proportional to the square of the distance from population concentrations. ReVelle, Cohon, and Shobrys (1991) adopted a multi-objective approach, trying to find an appropriate balance between minimization of transportation burden and perceived risk. Minimizing the economic aspects of the problem is a major concern in effectively managing waste management activities. In this regard, Current and Ratick (1995) studied a model with objectives related to cost, risk and equality in the distribution of the risk. In their model, the cost function is comprised of three cost terms; the first reflects the transportation cost and the second represents the cost associated with the fixed and variable costs of system facilities. In addition to considering environmental engineering and safety issues, minimization of the cost can also be observed in the works of Jacobs and Warmerdam (1994) and Cappanera, Gallo, and Maffioli (2003).

Nema and Gupta (1999) and Nema and Gupta (2003) addressed the problem of optimizing hazardous waste management system in the presence of treatment and disposal facilities. Two more

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https://doi.org/10.1016/j.ejor.2018.07.024
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realistic constraints on waste types properties, including waste-waste and waste-technology compatibility requirements, were introduced by the authors to provide insights for future research directions in this field. As a result, Alumur and Kara (2007) developed linear constraints to formulate the waste-technology compatibility constraint. According to their formulation, a waste can be treated using only its compatible technology. The second type of compatibility constraint has been taken into account in the formulation proposed by Rabbani, Heidari, Farrokhi-Asl, and Rahimi (2018). In their study, the waste-waste compatibility constraint ensures that a waste is not transported with its incompatible wastes i.e., incompatible waste types are collected separately employing a heterogeneous fleet of vehicle. Emek and Kara (2007) utilized an integer programming model for the location problem of disposal centers with treatment and recycling facilities. The government regulations and the air pollution standards play critical roles in their study. Samanlioglu (2013) and Boyer, Sai Hong, Pedram, Mohd Yusuff, and Zulkifli (2013) investigated a comprehensive waste management system composed of recycling, treatment and disposal facilities with all possible relationships between system components. Considering fuzzy service level requirements for waste generation centers, Ghezavati and Morakabatchian (2015) further extended the work of Samanlioglu (2013). In the field of the hazardous waste management, the most frequently employed approach to deal with the multi-objective optimization problems is the weighted sum method which is used for balancing costs, transportation and location risks. The first study that utilized an exact multi-objective solution approach (i.e., the lexicographic weighted Tchebycheff) to achieve a comprehensive Pareto optimal solutions for this class of problems was reported by Samanlioglu (2013). Later, Yu and Solvang (2016) and Zhao, Huang, Lee, and Peng (2016) applied the augmented epsilon-constraint method for this purpose.

It should be noted that optimizing the routing of hazardous materials in the mentioned studies, and most of other studies not discussed here, is based on a continuous network flow model. As one of the few research attempts focusing on vehicle routing problem, Zhao and Verter (2015) addressed the issue of establishing an optimization model for the configuration of a used-oil management system considering vehicle routing decisions. More recently, Rabbani et al. (2018) also incorporated the vehicle routing problem in designing a waste management system with multiple incompatible types of wastes. In addition to the facility location and vehicle routing decisions, the design of a logistic system consists of a third major category, namely the inventory control decision. Since such decisions are largely dependent, an integrated problem is required for real-life applications to outperform the traditional approaches that decompose decisions and often yield sub-optimal solutions (Drexl & Schneider, 2015). To this end, Javid and Azad (2010) innovated a logistic model to locate and allocate a set of facilities, determine their inventory policy and schedule vehicles' routes in a stochastic supply chain system. In fact, inventory control systems are essential for consistent availability of suppliers (Darwish & Odah, 2010) in order to guarantee service to processes or customers in a cost-effective manner (Huang, Tsai, Wu, & Chung, 2010). Therefore, in addition to cost considerations, maintaining customer service levels in terms of delivery date is an important concern in logistic systems (Nekooghadirli, Tavakkoli-Moghaddam, Ghezavati, & Javanmard, 2014). Nevertheless, in the hazardous waste management area, as a special class of reverse logistic, we no longer face the issue of customer satisfaction but face the risk issues. In this area, Tunalıoğlu, Koç, and Bektaş (2016) constituted the first effort to incorporate the inventory decisions into the location and routing problem in an olive oil mill wastewater management system with only treatment facilities. They assumed that treatment facilities can store a specified amount of working stock. Moreover, to minimize the total cost and environmental risks, Zhao and Ke (2017) incorporated the risk associated with the inventory of explosive wastes in a location-routing optimization model. The inventory decisions in their work are based on a single period framework and seek to maximize the transportation efficiency through an optimal utilization of vehicles capacity.

In all the previously cited studies, the location-routing problem was addressed under a deterministic environment ignoring the dynamic and transformational nature of the practical situations. Therefore, due to the severe effects of such uncertainties on planning decisions, not paying attention to the uncertain nature of the problem may contribute to sub-optimal or even infeasible design solutions (Nasiri, Heidari, Yazdanparast, & Akbarian, 2017). The importance of uncertainty has prompted a few researchers to focus on the uncertainty that can appear in various forms in location and routing-related problems in this area. Zhang and Zhao (2011) have investigated a problem focusing on waste generation amount uncertainties. The uncertain nature of this parameter in real-life situations stem from the complexity and fluctuations of the production processes in different industries. Transportation cost was considered as a source of uncertainty in the work of Ardjmand et al. (2016). It is obvious that with respect to the random capacity variations and different traffic conditions, transportation time or cost in urban road networks is strongly uncertain.

Location-routing is a prominent NP-hard Problem (Perl & Daskin, 1985), and therefore, due to its computational limits, exact solution approaches have found a more widespread acceptance to optimize only small-size versions of the problem. In addressing real-size problems, there is no hiding the fact that metaheuristic approaches have proven their applicability and efficiency (Lopez-Garcia, Onieva, Osaba, Masegosa, & Perallos, 2016). Recently, metaheuristic approaches have gained attention in dealing with the location-routing problem in the context of hazardous waste management. Ardjmand, Weckman, Park, Taherkhani, and Singh (2015) have developed a novel Genetic Algorithm (GA) to minimize the weighted sum of the cost and perceived risk in the problem of location and routing in facilities and disposal sites. Applying GA for a stochastic version of the mentioned problem is the topic of another study (Ardjmand et al., 2016) the obtained results of which indicated the superiority of the proposed GA algorithm in terms of yielding high-quality, non-dominated solutions and required CPU time. More recently, Asgari, Rajabi, Jamshidi, Khatami, and Farahani (2017) have developed an effective Memetic Algorithm (MA) combined with a Tabu Search (TS) algorithm to achieve near-optimal solutions in a reasonable time for a real-life case study.

Table 1 outlines a taxonomy of the relevant studies on hazardous waste management network design from 1989 to 2018 to overcome their deficiencies. The taxonomy contains five major categories of the model condition, some divided into sub-categories that further explain the model.

As can be found from our literature review, there are some more realistic aspects that have not been taken into consideration in the previous studies and could be expressed as follows:

- Simultaneous optimization of location, routing and inventory plans have not been conducted under uncertain conditions.
- Multi-period planning and uncertainty have not been incorporated into the formulation of a complex waste management system composed of treatment, recycling and disposal facilities.
- Application of a multi-objective solution approach to tackle the medium and large-scale stochastic versions of the problem has not been addressed in any of the previous studies.

This study is an effort to bridge the stated gaps with contributions in both model formulation and solution methodology. In terms of the model formulation, a stochastic multi-period model is developed, in which uncertainty is related to the amount of

Table 1
Taxonomy of the literature on the topic of hazardous waste management network design.

Study	System framework[a]				Compatibility[b]		Decisions[c]				Uncertainty	Multi-period
	T	S	R	D	W-W	W-T	L	A	R	I		
Zografros and Samara (1989)				*			*	*				
List and Mirchandani (1991)	*	*		*			*	*				
ReVelle et al. (1991)		*					*	*				
Current and Ratick (1995)	*			*			*	*				
Jacobs and Warmerdam (1994)		*		*			*	*				
Cappanera et al. (2003)				*			*	*				
Nema and Gupta (1999)	*			*	*	*	*	*				
Nema and Gupta (2003)	*			*	*	*	*	*				
Alumur and Kara (2007)	*			*		*	*	*				
Emek and Kara (2007)	*		*	*			*	*				
Samanlioglu (2013)	*		*	*		*	*	*				
Boyer et al. (2013)	*		*	*		*	*	*				
Ghezavati and Morakabatchian (2015)	*	*	*	*		*	*	*				
Yu and Solvang (2016)	*		*	*		*	*	*				
Zhao et al. (2016)	*		*	*		*	*	*				
Zhao and Verter (2015)	*			*		*	*	*	*			
Tunalıoğlu et al. (2016)	*						*	*	*	*		*
Zhang and Zhao (2011)	*			*		*	*	*			*	
Ardjmand et al. (2016)	*			*		*	*	*			*	
Ardjmand et al. (2015)	*			*		*	*	*				
Asgari et al. (2017)	*			*		*	*	*				
Zhao and Ke (2017)		*	*				*	*	*	*		
Rabbani et al. (2018)	*		*	*	*	*	*	*	*	*		
This study	*		*	*	*	*	*	*	*	*	*	*

[a] T: Treatment facility, R: Recycling facility, S: Storage facility, D: Disposal facility.
[b] W-W: waste-waste compatibility constraint, W-T: waste-technology compatibility constraint.
[c] L: Location, A: Allocation, R: Routing, I: Inventory.

generated waste and number of people at risk resulting from waste management activities. Moreover, the model is capable of handling three levels of decisions: location, inventory, and routing. The methodological contribution of this study is proposing a simulation-optimization approach based on a multi-objective evolutionary algorithm to solve a real-size optimization problem under uncertainty.

The rest of this paper is organized as follows. Following the introduction, a description of the problem and its formulation are presented in Section 2. Solution methodology is presented in Section 3. Model validation, parameters tuning and numerical results for several problem instances are reported in Section 4. Finally, Section 5 concludes the paper.

2. Problem description

The industrial hazardous waste management system in this study deals with all the treatment or handling processes from waste generation centers to final disposal. The system fits into the framework introduced by Rabbani et al. (2018), where different types of wastes are first collected from sources and then transported to the treatment or recycling facilities. After a proper treatment, a part of the waste stream can be fed into the recycling facilities, while the other generated residues must be directly sent to the landfill sites. Fig. 1 depicts the framework of the hazardous waste management system of this study.

The first key component of our optimization model for designing an effective waste management system is the vehicle routing problem. Because of the heterogeneous nature of the hazardous waste flow, waste-waste incompatibility constraints must be considered when scheduling a vehicle's route. In other words, for the purpose of avoiding chemical reactions, incompatible hazardous waste types are not permitted to be transported together. Moreover, the shipment of each waste type must have special requirements and for this reason, the system benefits from the advantages of a heterogeneous fleet of collection vehicles. At the beginning of each time period, a finite number of heterogeneous vehicles are available at a central depot from where they start their trip. After dispatching from the central depot, the vehicles pick-up wastes from a set of source nodes, dump their loads into the compatible treatment or recycling facilities, and finally terminate at the same depot. It is necessary to mention that each source must be visited only once by each type of collection vehicle.

The other component in our formulation to enhance the transport integration and to improve the system efficiency is the waste storage. Each treatment facility is designed in a way that it can hold a finite volume of waste as inventory by the end of the time horizon. In fact, an important feature of the treatment facilities in our study is their specific cost structure, where we face a relatively large fixed cost attached to their set up but a low variable cost associated with their operation. In our multi-period setting, the daily planning of each treatment facility must determine whether the total waste on that day should be processed or will be carried to the next time period as inventory. The inventory level in each treatment facility during different time periods is calculated based on Eq. (1).

$$\textit{Inventory level in perid } h = \begin{cases} 0; & \textit{if waste is treated in period } h \\ \left(\textit{Inventory level in perid } h - 1 \right. \\ \left. + \textit{ collected waste in period } h\right); & \textit{if waste is not treated in period } h \end{cases} \quad (1)$$

Therefore, the model aims at assisting decision makers in running treatment facilities at appropriate capacity with lower average costs. Furthermore, the generated wastes must be collected as soon as possible because longer collection time would result in contamination and storage costs which are linear in the quantity of uncollected waste at each time period.

The third component is associated with selecting the optimal locations of facilities among the available potential locations. In this problem, due to implementation costs and setup times, decisions concerning facility locations are quite stable in time. Consequently, the locational decisions are excluded from the finite

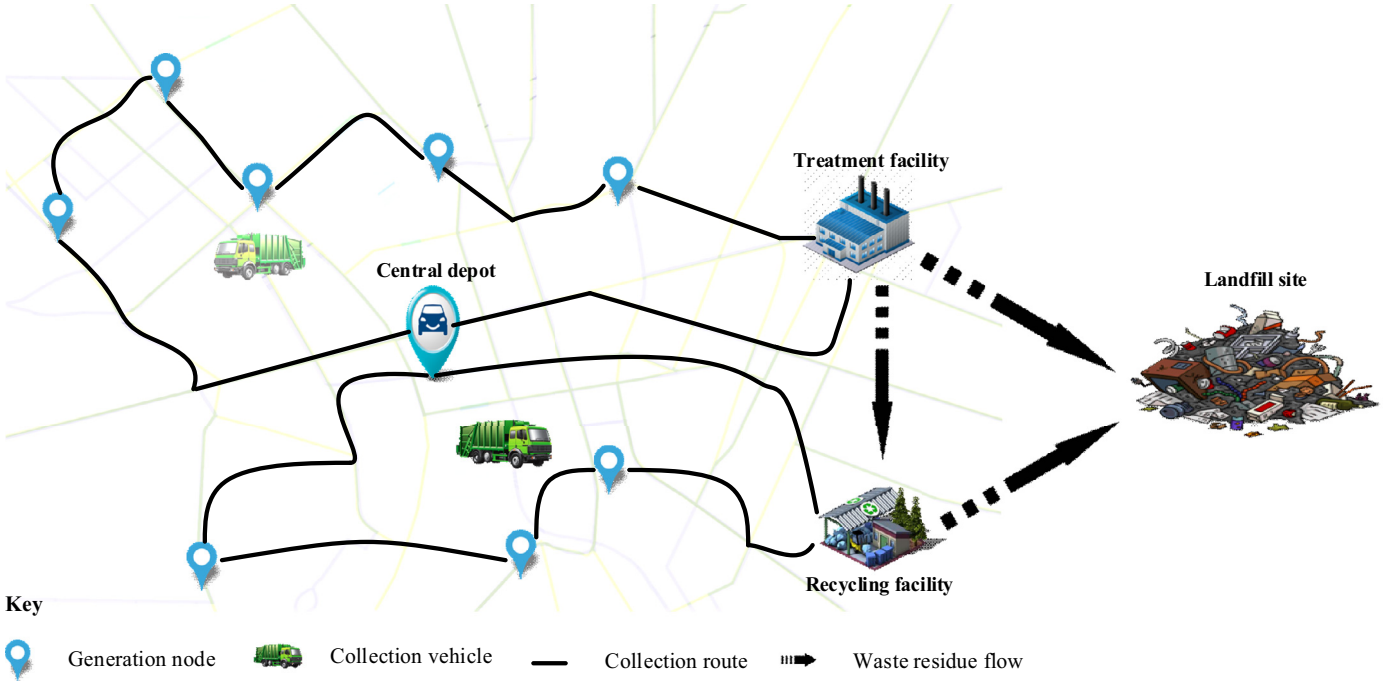


Fig. 1. Framework of the hazardous waste management system.

decisions, including routing and inventory decisions. In other words, these two kinds of decisions are made at different time scales. In this component, locational decisions are made only once at the beginning of the first time horizon and remain unchanged during all other planning horizons. Bear in mind that the problem is defined over a discrete setting and a week planning horizon divided into days.

The formulation of our model which aims to simultaneously minimize the total cost and the total environmental risk imposed on nearby residents is based on some assumptions given as follows:

- All the network design decisions are implemented at the beginning or the end of the time periods.
- Each treatment facility can be equipped only with one of the various available technologies.
- In case of some treatment technologies, the system suffers from high fixed setup costs.
- Uncollected waste at generation nodes and unprocessed waste at treatment centers in each time period impose storage costs on the system.
- None of the facilities has working stock at the beginning or end of the time horizon.
- The risk associated with facilities is linearly related to the amount of hazardous waste or waste residues available at each facility and the number of people within a certain distance of it.
- The risk attributes along different transportation links are linearly related to the waste amount transported and the number of people along a certain distance of routes.
- Amounts of waste generated from different sources as well as the population at risk are assumed to be stochastic parameters.
- There exist a given number of collection vehicles in the central depot.
- Realizations of the amount of waste generated at different sources do not exceed the capacity of vehicles.

Given these assumptions, we integrate the aforementioned components in a mathematical programming model in the next sub-section.

2.1. Mathematical model

In this section, we propose a multi-objective Mixed Integer Non-Linear Programming (MINLP) model for the problem. Regarding the basic assumptions given above, this model is considered a useful decision-making tool for real-world applications. Before we embark on modeling our problem, notations which will be used to model the problem are introduced in Eqs. (2)–(32). Note that the stochastic parameters are represented by the $\sim$ symbol above them.

Index set

G	source nodes; indexed by g
T	pool of existing and potential treatment facilities; indexed by t
$T' \subset T$	existing treatment facilities; indexed by t'
R	pool of existing and potential recycling facilities; indexed by r
$R' \subset R$	existing recycling facilities; indexed by r'
F	pool of existing and potential disposal facilities; indexed by f
$F' \subset F$	existing disposal facilities; indexed by f'
Q	treatment technologies; indexed by q
D	central depot; indexed by d
W	hazardous waste types; indexed by w
K	fleet of collection vehicles; indexed by k
H	time periods; indexed by h

Parameters

c_{ij}	transportation cost of one unit of waste between node i and j
ft_{qi}	establishment cost of a treatment facility with technology q at location $i \in (T - T')$
fr_i	establishment cost of a recycling facility at location $i \in (R - R')$

ff_i establishment cost of a disposal facility at location $i \in (F - F')$
$\widetilde{pop_{ij}}$ number of people are affected by transportation risk between node i and j
$\widetilde{PA_i}$ number of people are affected by the risk of the facility at location $i \in (T \cup F)$
$\widetilde{d_{gw}}$ amount of waste type w accumulated at the source node g
β_{wq} percentage of recyclable waste residues produced during processing of waste type w with technology q
r_{wq} percentage of mass reduction during processing of waste type w with technology q
ω_r percentage of total waste recycled in recycling facility r
tc_t capacity of treatment facility t
rc_r capacity of recycling facility r
fc_f capacity of disposal facility f
δ_w capacity of a vehicle that is compatible with waste type w
fct_{qt} fixed setup cost of technology q in treatment facility t
vct_{qt} variable cost associated with processing one unit of waste with technology q in treatment facility t
vcr_r variable cost associated with recycling one unit of waste in recycling facility r
vcf_f variable cost associated with disposing one unit of waste in disposal facility f
hc_i storage cost of one unit of waste at node $i \in (G \cup T \cup R \cup F)$
com_{wq} binary parameter (1 if waste type w is compatible with treatment technology q; 0 otherwise)
v_{wk} binary parameter (1 if waste type w is compatible with vehicle k; 0 otherwise)
$a_{qt'}$ binary parameter (1 if existing treatment facility t' is equipped with technology q; 0 otherwise)

Decision variable

x^h_{ijk} binary variable (1 if node j is visited just after node i by vehicle k in period h; 0 otherwise)
t_{qi} binary variable (1 if treatment technology q is established at node $i \in (T - T')$; 0 otherwise)
r_r binary variable (1 if a recycling facility is established at node r; 0 otherwise)
f_f binary variable (1 if a disposal facility is established at node f; 0 otherwise)
pt^h_{qt} binary variable (1 if accumulated wastes at location t are treated with technology q in period h; 0 otherwise)
l^h_{ik} load of vehicle k after visiting node i in period h
xt^h_{wt} amount of waste type w transferred to treatment facility t in period h
xr^h_r amount of recyclable waste transferred to recycling facility r in period h
xf^h_f amount of waste transferred to disposal facility f in period h
xtr^h_{tr} amount of recyclable waste residue transported from treatment facility t to recycling facility r in period h
xtf^h_{tf} amount of waste residue transported from treatment facility t to disposal facility f in period h
xrf^h_{rf} amount of waste residue transported from recycling facility r to disposal facility f in period h
st^h_{wt} amount of working stock of waste type w in treatment facility t in the end of period h

2.1.1. Objective functions

The model considers three main objective functions to minimize the total cost and the environmental risk. The first objective consists of five components related to the collection, transportation, investment and operating cost of facilities, and waste storage cost in Eqs. (2.1)–(2.5), respectively. This objective aims at minimizing the total cost.

$$Min\, f_1(x) = \sum_{h\in H}\sum_{k\in K}\sum_{i\in G}\sum_{j\in G\cup T\cup R} c_{ij}\, x^h_{ijk}\, l^h_{ik} \tag{2.1}$$

$$+ \sum_{h\in H}\sum_{i\in T}\sum_{j\in R} c_{ij}\, xtr^h_{ij} + \sum_{h\in H}\sum_{i\in T}\sum_{j\in F} c_{ij}\, xtf^h_{ij} + \sum_{h\in H}\sum_{i\in R}\sum_{j\in F} c_{ij}\, xrf^h_{ij} \tag{2.2}$$

$$+ \sum_{q\in Q}\sum_{i\in T-T'} ft_{qi}\, t_{qi} + \sum_{i\in R-R'} fr_i\, r_i + \sum_{i\in F-F'} ff_i\, f_i \tag{2.3}$$

$$+ \sum_{h\in H}\sum_{w\in W}\sum_{q\in Q}\sum_{i\in T} pt^h_{qi}\left(fct_{qi} + vct_{qi}\left(st^{h-1}_{wi} + xt^h_{wi}\right)\right) + \sum_{h\in H}\sum_{i\in R} vcr_i\, xr^h_i + \sum_{h\in H}\sum_{i\in F} vcf_i\, xf^h_i \tag{2.4}$$

$$+ \sum_{h\in H}\sum_{i\in G} hc_i\left(\sum_{w\in W}\widetilde{d_{iw}} - \sum_{h'=1}^{h}\sum_{k\in K}\sum_{w\in W}\sum_{j\in D\cup G} x^{h'}_{jik}\,\widetilde{d_{iw}}\, v_{wk}\right) + \sum_{h\in H}\sum_{w\in W}\sum_{q\in Q}\sum_{i\in T} hc_i\left(1 - pt^h_{qi}\right)\left(st^{h-1}_{wi} + xt^h_{wi}\right) \tag{2.5}$$

The second objective, given in Eqs. (3.1) and (3.2), computes the total site risk that is a function of the waste quantity being processed at treatment and disposal facilities.

$$Min f_2(x) = \sum_{h\in H}\sum_{w\in W}\sum_{i\in T}\widetilde{PA_i}\left(st^{h-1}_{wi} + xt^h_{wi}\right) \tag{3.1}$$

$$+ \sum_{h\in H}\sum_{i\in F}\widetilde{PA_j}\, xf^h_i \tag{3.2}$$

Eqs. (4.1) and (4.2) form the last objective function, and are designed to minimize the transportation risk for the people present on and along all possible routes.

$$Min\, f_3(x) = \sum_{h\in H}\sum_{k\in K}\sum_{i\in G}\sum_{j\in G\cup T}\widetilde{pop_{ij}}\, x^h_{ijk}\, l^h_{ik} \tag{4.1}$$

$$+ \sum_{h\in H}\sum_{i\in T}\sum_{j\in F}\widetilde{pop_{ij}}\, xt f^h_{ij} \tag{4.2}$$

2.1.2. Constraints

At the beginning of each time period, all collection vehicles are located in a central depot. According to Eq. (5), the central depot is the initial position of all collection vehicles.

$$\sum_{i\in D}\sum_{j\in G} x^h_{ijk} = 1;\ \forall k \in K,\ \ h \in H \tag{5}$$

The continuity of collection routes results from Eq. (6). Under this constraint, if a collection vehicle enters a node, it must also leave the node to another destination.

$$\sum_{i\in D\cup G} x^h_{ijk} - \sum_{i'\in G\cup R\cup T} x^h_{ji'k} = 0;\ \forall j \in G, k \in K, h \in H \tag{6}$$

An underlying assumption in our formulation is that the split pickup and delivery are not permitted. Given this assumption, Eq. (7) guarantees that each source node is visited only once by each type of vehicle.

$$\sum_{h\in H}\sum_{k\in K}\sum_{j\in G\cup R\cup T} x^h_{ijk}\, v_{kw} = 1;\ \forall w \in W, i \in G \tag{7}$$

Setting the end point of collection vehicles on the same central depot is done by adding Eq. (8) to the model.

$$\sum_{i\in G} x^h_{ijk} - \sum_{i'\in D} x^h_{ji'k} = 0;\ \forall k \in K, \mathrm{j} \in R \cup T, h \in H \tag{8}$$

In addition to the waste-waste incompatibility constraint, there exists another incompatibility constraint. This indicates that different technologies are used to deal with different types of non-recyclable hazardous wastes. Eq. (9) enforces such constraints when scheduling collection routes.

$$x^h_{ijk} \le \sum_{w\in W}\sum_{q\in Q} v_{kw}\, \mathrm{com}_{wq}\, \mathrm{t}_{qj};\ \forall k \in K, \mathrm{i} \in G, j \in T, h \in H \tag{9}$$

Based on Eq. (10), all collection vehicles with recyclable loads have to deliver their loads to recycling facilities. To simplify the formulation of this constraint, we assume that recyclable waste type is not compatible with any treatment technology.

$$x^h_{ijk} \le \sum_{w\in W} v_{kw}\, \frac{((|Q| - \sum_{q\in Q} com_{wq})\, r_j)}{|Q|};\ \forall k \in K, \mathrm{i} \in G, j \in R, h \in H \tag{10}$$

Eqs. (11)–(14) guarantee two important properties in our formulation: eliminating sub-tour and not exceeding the capacity of vehicles.

$$l^h_{ik} - l^h_{jk} + \sum_{w\in W} v_{kw}\, \delta_w\, x^h_{ijk} \le \sum_{w\in W} v_{kw}\, (\delta_w - \widetilde{\mathrm{d}_{jw}});\ \forall i, j \in G, k \in K, h \in H \tag{11}$$

$$\sum_{w\in W} \widetilde{d_{iw}}\, v_{kw} \le l^h_{ik} \le \sum_{w\in W} v_{kw}\, \delta_w;\ \forall \mathrm{i} \in G, k \in K, h \in H \tag{12}$$

$$x^h_{ijk}\left(\sum_{w\in W} \widetilde{d_{jw}}\, \mathrm{v}_{kw}\right) \le l^h_{jk};\ \forall i \in D,\ \ j \in G, k \in K, h \in H \tag{13}$$

$$l^h_{jk} \le \sum_{w\in W} v_{kw}\left(\delta_w + \left(\sum_{i\in D} \widetilde{d_{jw}} - \delta_w\right) x^h_{ijk}\right);\ \forall j \in G, k \in K, h \in H \tag{14}$$

The non-recyclable waste amounts to be delivered to the target treatment facilities are determined using Eq. (15).

$$xt^h_{wj} = \sum_{k\in K}\sum_{i\in G} x^h_{ijk}\, l^h_{ik}\, \mathrm{v}_{kw};\ \forall j \in T, w \in W, h \in H \tag{15}$$

After treatment processes, a part of the waste residues can be sent to the recycling facilities. Eq. (16) determines the amount of flow from treatment centers to recycling facilities in each time period.

$$\sum_{w\in W}\sum_{q\in Q} pt^h_{qi}\left(st^{h-1}_{wi} + xt^h_{wi}\right)(1 - r_{wq})\,\beta_{wq} = \sum_{j\in R} xtr^h_{ij};\ \forall i \in T, h \in H \tag{16}$$

The amount of waste to be recycled in different recycling facilities is determined using Eq. (17).

$$xr^h_j = \sum_{w\in W}\sum_{k\in K}\sum_{i\in G} x^h_{ijk}\, l^h_{ik}\, v_{kw} + \sum_{i'\in K} xtr^h_{i'j};\ \forall j \in R, h \in H \tag{17}$$

The non-recyclable waste residues produced in different facilities form the input flow to the disposal facilities. At first, Eqs. (18) and (19) calculate the amount of waste residues that must be directly shipped to the disposal facilities from treatment and recycling facilities, respectively. Then, Eq. (20) determines the amount of waste, which must be disposed in corresponding facilities by summation of these two streams.

$$\sum_{w\in W}\sum_{q\in Q} pt^h_{qi}\left(st^{h-1}_{wi} + xt^h_{wi}\right)(1 - r_{wq})\,(1 - \beta_{wq}) = \sum_{j\in F} xtf^h_{ij};\ \forall i \in T, h \in H \tag{18}$$

$$xr^h_i \times (1 - \omega_i) = \sum_{j\in F} xrf^h_{ij};\ \forall i \in R, h \in H \tag{19}$$

$$xf^h_i = \sum_{j\in T} xtf^h_{ji} + \sum_{j\in R} xrf^h_{ji};\ \forall i \in F, h \in H \tag{20}$$

At the end of each time period, the level of working stock in different treatment facilities is updated based on Eqs. (21) and (22).

$$st^h_{wi} = \left(1 - \sum_{q\in Q} pt^h_{qi}\right)\left(st^{h-1}_{wi} + xt^h_{wi}\right);\ \forall i \in T, w \in W, h \in H \tag{21}$$

$$pt^h_{qi} \le t_{qi};\ \forall i \in T, \mathrm{q} \in Q,\ \ h \in H \tag{22}$$

The existing treatment, recycling, and disposal facilities are specified in Eqs. (23)–(25), respectively. Furthermore, the maximum number of treatment technologies that can be applied to a selected candidate location is restricted using Eq. (26).

$$t_{qi} = a_{qi};\ \forall i \in T',\ \ q \in Q \tag{23}$$

$$r_i = 1;\ \forall i \in R' \tag{24}$$

$$f_i = 1;\ \forall i \in F' \tag{25}$$

$$\sum_{q\in Q} t_{qi} \le 1;\ \forall i \in T \tag{26}$$

No working stock exists at any of the treatment centers at the beginning or end of the time horizon, this constraint is equivalent to Eq. (27).

$$st^0_{wi} + st^{\max\{H\}}_{wi} = 0;\ \forall i \in T, w \in W \tag{27}$$

The capacity limitations on treatment, recycling, and disposal facilities are satisfied using Eqs. (28), (29) and (30), respectively.

$$\sum_{w\in W} st^{h-1}_{wi} + xt^h_{wi} \le \mathrm{tc}_i \sum_{q\in Q} \mathrm{t}_{qj};\ \forall i \in T, h \in H \tag{28}$$

$$xr^h_i \le rc_i\, r_i;\ \forall i \in R, h \in H \tag{29}$$

$$xf^h_i \le fc_i\, f_i;\ \forall i \in F, h \in H \tag{30}$$

Constraints on decision variables are represented by Eqs. (31) and (32).

$$\begin{gathered} xtf^h_{ij} \ge 0, xf^h_j \ge 0;\ \forall i \in T, j \in F, h \in H \\ xtr^h_{ij} \ge 0, xt^h_{wi} \ge 0, st^h_{wi} \ge 0;\ \forall i \in T, j \in R, w \in W, h \in H \\ l^h_{ik} \ge 0;\ \forall i \in G, k \in K, h \in H \end{gathered} \tag{31}$$

$$\begin{gathered} x^h_{ijk} \in \{0,1\};\ \forall i, j \in (D \cup G \cup R \cup T), k \in K, h \in H \\ r_i \in \{0,1\};\ \forall i \in R \\ f_i \in \{0,1\};\ \forall i \in F \\ t_{qi}, pt^h_{qi} \in \{0,1\};\ \forall i \in T, q \in Q, h \in H \end{gathered} \tag{32}$$

2.2. Model linearization

The developed model is initially a MINLP model that will be linearized in this sub-section using an exact linearization method (Azadeh, Shafiee, Yazdanparast, Heydari, & Keshvarparast, 2017). The non-linearity of the model comes from the presence of cross-product terms in the structure of the model. All non-linear terms, including $x^h_{ijk} l^h_{ik}$, $pt^h_{qi} st^{h-1}_{wi}$ and $pt^h_{qi} xt^h_{wi}$ are the cross-products of a binary variable and a continuous variable. A simple method to evade such non-linearities is defining a new continuous variable and three auxiliary constraints in the model. In this regard, we define a continuous variable xl^h_{ijk} and replace the non-linear term $x^h_{ijk} l^h_{ik}$. In order to ensure that this reformulation will yield the same results as the initial non-linear formulation, Eqs. (33)–(35) must be added to the model. These Equations are as follows:

$$xl^h_{ijk} \le BM\, x^h_{ijk}; \forall i \in (D \cup G), \mathrm{j} \in G,\ k \in K, h \in H \tag{33}$$

$$xl^h_{ijk} \le l^h_{jk}; \forall i \in (D \cup G), \mathrm{j} \in G,\ k \in K, h \in H \tag{34}$$

$$xl^h_{ijk} \ge l^h_{jk} - \left(1 - x^h_{ijk}\right) BM; \forall i \in (D \cup G), \mathrm{j} \in G,\ k \in K, h \in H \tag{35}$$

By applying the same approach, the non-linearities associated with $pt^h_{qi} st^{h-1}_{wi}$ and $pt^h_{qi} xt^h_{wi}$ will be overcome, and the initial non-linear formulation will be reduced to a MILP model.

3. Solution methodology

The use of metaheuristic approaches in solving many difficult problems has rapidly gained popularity in a wide variety of disciplines. However, the high degree of uncertainty that reflects realistic problem settings in practice contributes to the less effectiveness of single metaheuristics (Dinh & Mamun, 2004). Simulation-optimization algorithms have widely been applied to deal with complex, large scale, and dynamic real-world scenarios under uncertainty conditions (Juan, Rabe, Faulin, & Grasman, 2015). As a growing research field inside the simulation-optimization area, simheuristic approaches seek to fill this lack by incorporating simulation into a (meta-) heuristic-driven framework (Juan, Faulin, Grasman, Rabe, & Figueira, 2015). Simheuristics can be used in solving the combinatorial optimization problems with stochastic components in the objective functions and the set of constraints that are not easy to be captured in reasonably short computing times. In fact, hybrid simulation-optimization techniques are used to improve existing heuristics and enable them to contribute to the improved efficiency (Juan et al., 2011b).

The tremendous advancement in computer processing power and in developing pseudo-random number generators has offered new prospects in utilizing probabilistic approaches such as Monte Carlo simulation in solving combinatorial problems (Juan, Faulin, Ruiz, Barrios, & Caballé, 2010). Monte Carlo simulation can be expressed as a set of techniques that make use of random numbers and probability distributions in handling certain stochastic and deterministic problems (Law, Kelton, & Kelton, 2007). When suitably integrated with heuristic methods, Monte Carlo simulation has proven to be advantageous in solving stochastic combinatorial optimization problems (Juan et al., 2011a). Unlike deterministic problems, a major challenge in the face of stochastic problems is the stochastic nature of the fitness value evaluation. This simulation technique overcomes this difficulty by utilizing a given number of samples in which the true fitness value of the solution is estimated by its unbiased estimator, i.e. the sample mean of the fitness values (Zhang et al., 2017). Another approach to select the best solution is conducting statistical tests for comparing the sample average values of the solutions.

Monte Carlo simulation is extensively employed for addressing probabilistic uncertainty in most simheuristic approaches (Kenyon & Morton, 2003). In simulation-based approaches, Monte Carlo simulation can be used to estimate the stochastic behavior of the solutions and their reliability (see, for instance, Juan et al. (2011a) and Wang and Lin (2013)). In addition, the feedback of the simulation part is fed to the approach to guide the stochastic search process (de Armas, Juan, Marquès, & Pedroso, 2017; Gruler, Fikar, Juan, Hirsch, & Contreras-Bolton, 2017). In real-life problem settings like the waste management problems, there exist reliable experience values to obtain good estimates of the key parameters. Therefore, Monte Carlo simulation can be used to simulate such parameters and obtain realistic and reliable solutions (Sonnemann, Schuhmacher, & Castells, 2003). As an example in the area of waste management, the methodology developed by Gruler et al. (2017) embedded a Monte Carlo simulation within a metaheuristic framework in order to estimate the expected cost and cost variability of promising routing plans, which led to reliable routing solutions.

To the best of our knowledge, there exists no previous work concerning such stochastic approaches in the multi-objective combinatorial optimization literature. In this regard, a scenario-based optimization approach i.e., the application of Monte Carlo sampling on a multi-objective metaheuristic approach, is developed in this paper to solve the proposed problem. The remaining of this section is devoted to a brief description of the proposed simheuristic approach and how to encode individual solutions by it.

3.1. The proposed simheuristic approach

A property that characterizes our problem is the discontinuous nature of solution space. The application of genetic algorithms in handling discrete design variables is relatively clear since they can directly handle the discrete nature of the problem (Martínez-Salazar, Molina, Ángel-Bello, Gómez, & Caballero, 2014). Another relevant feature that characterizes this problem is the existence of multiple conflicting objectives. In tackling such problems, the primary advantages of NSGA-II, as the multi-objective variant of genetic algorithm, are its low time complexity (Azadeh, Shafiee, Yazdanparast, Heydari, & Fathabad, 2017) as well as its ability in finding a diverse pool of solutions for problems with non-convex, non-smooth and multi-modal solution spaces (Goyal, Jain, & Jain, 2012). For these reasons, NSGA-II is utilized as the foundation for the proposed simheuristic approach. The following steps explain how to implement the simheuristic approach.

Step 1: Generate an initial population of artificial chromosomes in the form of two-dimensional matrices with $W+1$ rows and G columns in which all cells are encoded with real numbers from the [0, 1] interval. Using this structure, we are able to capture simultaneously both the strategic and operational levels of our problem.

Step 2: Generate a certain number of scenarios, which are possible realizations of the stochastic parameters. The number of scenarios needed to appropriately reflect the uncertainty involved can be calculated using Eq. (36) (Brandimarte, 2014).

$$n = \left[z_{a/2} \frac{S}{E} \right]^2 \tag{36}$$

In this formula, n, S and E denote the number of scenarios, estimated standard deviation of the output and desired margin of error, respectively. Furthermore, $z_{a/2}$ represents the critical value of the normal distribution for $a/2$.

Step 3: Evaluate the multi-objective fitness for each individual in the population. To this end, it is necessary to extract the decision variables of the problem from the proposed solution structure. Location and routing decisions are made based upon the first W rows of the solution matrix in which each row is associated with a

waste type. For example, to construct the collection routes of waste type one during planning horizon, the cells in the first row are ranked in descending order of their value. The first route associated with this waste type starts by servicing the first-ranked generation node after the central depot. Then, high-ranked generation nodes are added to the current route, respectively, until adding another one will exceed the vehicle's capacity. Afterward, the nearest treatment or recycling facility (based on the waste type) with sufficient capacity is chosen for unloading the collected waste. If there exists no such facility in the network, a new facility in the nearest location is established. Finally, the route is terminated at the central depot and new routes will be started to service the remaining generation nodes. After determining the collection routes, it is necessary to assign the vehicles to the constructed routes. With respect to the limitation on the vehicle fleet size, it is evident that these routes should be allocated to the different time periods. This procedure is repeated for other waste types using rows 2, 3, …, and W. It is necessary to note that transportation routes are determined by minimizing the flows transit distance. In addition, our solution structure consists of a row to determine the inventory control decisions in system facilities. The last row of the matrix is designed for this purpose. To determine time periods in which the waste processing must be carried out, we transform the encoded values of this row to binary variables (0 or 1). The cell position in the row denotes the time period, the value greater or equal to 0.5 means the waste processing operations should be performed and the value smaller than 0.5 expresses there is no processing on the waste and the waste should be carried to the next time period. Regardless of the cell value, the waste processing operations are performed in the last time period as well as other periods in which inventory holding leads to an exceeding in the capacity of the facility. To sum up, the solution representation can determine the strategic location decisions, collection routes, transportation routes and the inventory holding policy in time periods as well. Once this is done, it is easy to evaluate the value of the objective function under different scenarios.

Step 4: Perform non-dominated sorting based on ranking and crowding distance. With respect to the evaluation of individual solutions under different scenarios and consequently existing a one-to-one correspondence between the objective function's values of each pair of solutions, the paired t-test is used in ranking solutions. By defining $D_{im} = f_{ai}^m - f_{bi}^m$ as the difference between objective function m for solutions a and b under scenario i ($i = 1, 2, \ldots, n$) and $\overline{D_m}$and S_{D_m} as the mean and standard deviation of these differences, solution a dominates b if the following two conditions are satisfied in our minimization problem:

The null hypothesis of the first paired t-test (i.e., $H_0 : \frac{\overline{D_m}-0}{S_{D_m}/\sqrt{n}} \leq 0$) is accepted for all objective functions at the five percent level of significance.

- There exists at least one objective function that the null hypothesis of the second paired t-test ($H_0 : \frac{\overline{D_m}-0}{S_{D_m}/\sqrt{n}} < 0$) is accepted at the five percent level of significance.

After checking these conditions for all pair of the solutions, solutions are sorted into different front levels based on the domination of pair comparison. Level 1 is the top level in which the solution is dominated by no other solution; level 2 is the second level in which the solution is dominated by some solutions only in level 1, and so on.

Considering the fact that a large diversity of the solutions can prevent results from trapping into a local optimum, it is more favorable that the location of the solutions in each front is distributed uniformly. Therefore, another feature, called crowding distance, is applied to measure how close an individual is to its neighbors. Another difference between the proposed simheuristic approach and the base algorithm is the method used to calculate this measure. In this study, initially the crowding distance associated with each solution is calculated under different scenarios. Then, for each solution, the obtained values are averaged.

Step 5: Apply selection, crossover, and mutation to create offspring population. To select parents in producing offspring, binary tournament selection which selects a solution from a Pareto front with a higher rank is applied. If the rank values of solutions are the same, then the solution with bigger crowding distance is selected. Afterwards, offspring population is formed using crossover and mutation operators on the selected population. Among various crossover and mutation techniques that can be applied, single-point crossover and inversion mutation are employed in this study. Random selection of a point and exchanging the portions of the two solutions beyond this point and also a random selection of two points and reversion of the columns between them reflect these types of crossover and mutation operators, respectively.

Step 6: Perform Step 3 for the offspring population.

Step 7: Perform Step 4 for the temporary population that is the union of the current generation and offspring populations.

Step 8: Select the next generation population. In a subsequent manner, the next generation is filled with the highest-ranked Pareto front solutions until it reaches the population size.

Step 9: Determine the most preferred solution among the Pareto optimal set. After the algorithm has converged to the Pareto optimal solutions, we rank the obtained solutions using TOPSIS method (Sun & Lin, 2009) by considering the objective functions as its criteria.

After providing a step-by-step explanation of our proposed simheuristic approach, it seems useful to employ the flowchart of Fig. 2 to summarize the method. As can be found from the flowchart, the distinguishing features of the proposed algorithm are highlighted with a different color. It is necessary to note that the stopping criteria in the algorithm is based on the generation limit.

In addition to the proposed solution approach, we use the hybrid simulation-analytical modeling approach introduced by Acar, Kadipasaoglu, and Day (2009) to validate the algorithm and its results for the stochastic multi-objective optimization problem of this study. Appendix A provides a brief description of this approach.

4. Results and discussion

4.1. Problem instances

To the best of our knowledge, there are no standard benchmark dataset for the extent of the stochastic optimization problem of this study as of now. Accordingly, we need to test the proposed solution approach and compare its obtained results with the ones provided by the hybrid simulation-analytical modeling approach using randomly generated test problems. The generated problems dimensions are presented in Table 2, which are categorized into small, medium and large-scale problems.

We assume, for simplicity, there are four types of waste in the waste management network of all problem instances. Furthermore, similar to Rabbani et al. (2018), we consider only incineration and chemical treatment as the available treatment technologies. A recyclable waste type, two non-recyclable waste types that are compatible with only one type of treatment technology and a waste type whose compatibility with both technologies is its main feature form the waste stream. Considering Table 2 data, we will randomly generate the parameters of the problem based on the uniform probability distribution functions provided in Appendix B. In addition, three levels of uncertainty regarding the stochastic parameters are considered by varying the standard deviations of the

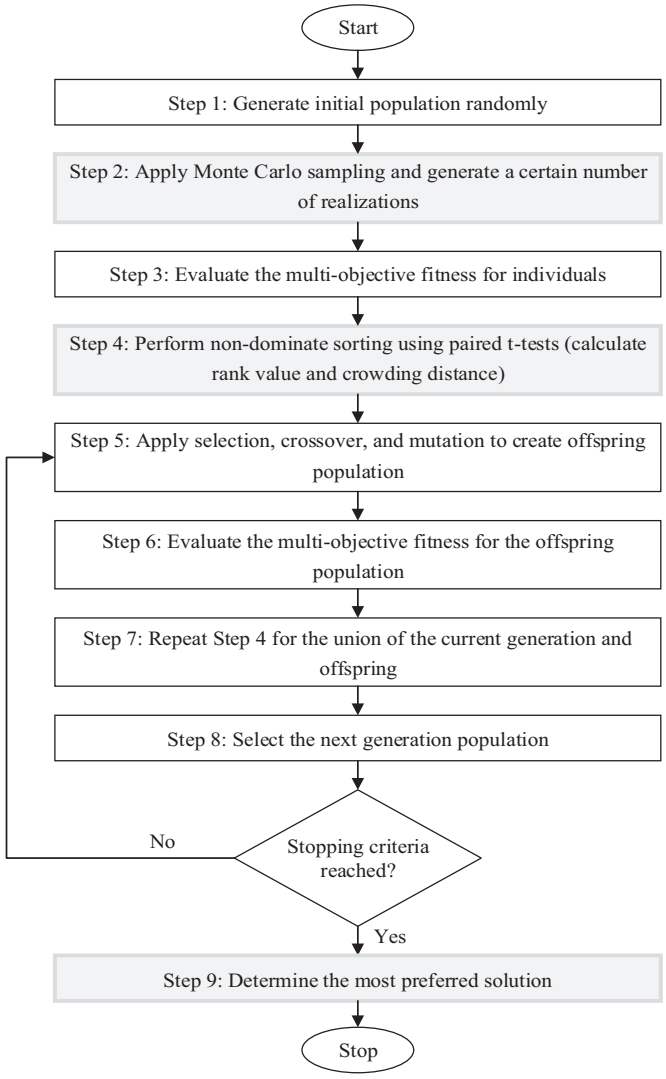


Fig. 2. Flowchart of the proposed simheuristic algorithm.

Table 2
Dimension of test problems.

Problem dimension	Problem no.	G	(T', T)	(R', R)	(F', F)	H
Small	1	6	(2, 4)	(2,3)	(2,4)	3
	2	8	(2, 4)	(2,3)	(2,4)	4
	3	10	(3, 5)	(2,3)	(2,4)	5
	4	12	(3, 5)	(2,3)	(2,4)	4
	5	15	(3, 5)	(2,4)	(2,4)	5
Medium	6	20	(3, 6)	(2,4)	(3, 5)	4
	7	20	(3, 6)	(2,4)	(3, 5)	4
	8	25	(3, 6)	(2,4)	(3, 5)	5
	9	25	(3, 6)	(2,4)	(3, 5)	5
	10	30	(3, 6)	(2,4)	(3, 5)	5
Large	11	40	(3, 7)	(2,4)	(3, 5)	5
	12	50	(3, 7)	(3, 5)	(3, 6)	5
	13	75	(3, 7)	(3, 5)	(3, 6)	5
	14	90	(4, 8)	(3, 5)	(4, 7)	5
	15	100	(4, 8)	(3, 5)	(4, 7)	5

parameters to 5% (low uncertainty), 10% (moderate uncertainty) and 25% (high uncertainty) of the mean value of parameters. We continue this section by reporting our computational experience on the test problems for both solution approaches.

Table 3
Optimal objective functions value for problem instance 1.

Objective function	Minimum value
Cost	36,829
Transportation risk	5,884.171
Site risk	51,596,860

4.2. Small-size problem instances

The test problems are used to illustrate the applicability of the developed model and the proposed solution approach for solving the optimization problem of this study. Before providing and discussing the results for all 15 instances under three uncertainty levels, we will use one in order to better illustrate our approach. To this end, first, problem instance 1 with moderate uncertainty level is treated in detail, and then the obtained results will be reported for all instances. It is worth noting that this process is conducted using GAMS and MATLAB software on an Intel Corei7 PC with 8 gigabytes of RAM and over 2 gigahertz CPU.

We begin the solution procedure by solving three deterministic (considering mean value of uncertain parameters) single objective problems, in each of which the objective function is considered as a single objective for the problem and their minimum values are calculated in the absence of other objectives. Table 3 provides the optimal value of objective functions of problem instance 1.

After determining the weight coefficients of objectives and converting the problem into a single objective one, we solve the problem without considering the uncertainty of the input parameters. To evaluate the effect of uncertainty, we simulate the obtained solution structure using a simple Excel macro written in Microsoft Visual Basic. In this regard, the results of different simulation runs in terms of objective function values are gathered and averaged. The process of solving the deterministic model and simulating its results is iterated until the optimum solution is achieved. Table 4 summarizes the results of hybrid simulation-analytical modeling approach during different iterations.

As can be understood from Table 4, the solution of iteration 2 outperforms the initial solution when the uncertainty impact is considered. Therefore, with respect to the resulting improvement in the solution, it is necessary to update the mathematical model to incorporate its uncertainty impact in the model during the future iterations. Similar to iteration 2, the updating process is repeated in iteration 3. During the following iterations, none of the resulting solutions can improve the best known solution (i.e., the solution of iteration 3) and the algorithm is terminated in iteration 10 when it reaches a repeated solution. Figs. 3 and 4 depict the trend of changes in the normalized weighted sum of objectives values for both without and with uncertainty impacts, respectively.

According to Figs. 3 and 4, the hybrid simulation-analytical modeling approach looks for a solution with the least value of the objective function under uncertainty. In other words, while the objective value in the deterministic model is increased during the consecutive iterations, this approach follows the path to find a solution with a minimum value of the objective function by considering the uncertainty impacts of the input parameters.

The obtained results from solving problem instance 1 reveal that incorporating uncertainty impact into decision making can lead to a 1.78 percent cost saving and increase the transportation and site risk to 1.5% and 0.96%, respectively. In the following, we report the results observed from solving the small-scale problems in terms of uncertainty impact on the objective functions. In Fig. 5, the average percent cost saving (shown in green color) along with the average percent increase of the risk measures (shown in light and dark red colors) are presented under different uncertainty

Table 4
Results of the hybrid simulation-analytical modeling approach for problem instance 1.

Iteration	Deterministic model				Simulation			
	Obj. 1	Obj. 2	Obj. 3	NWSO*	Obj. 1	Obj. 2	Obj. 3	NWSO*
1	45,242	6,225,509	79,827,910	1.266	48,861	6,723,549	86,214,142	1.367
2	47,391	6,231,983	80,709,120	1.299	50,234	6,543,582	84,341,030	1.363
3	46,135	6,539,817	83,431,610	1.308	47,989	6,827,568	87,044,198	1.360
4	46,033	6,308,717	87,225,120	1.316	47,874	6,567,374	90,626,899	1.368
5	47,952	6,329,019	84,063,450	1.327	49,870	6,588,508	87,594,114	1.380
6	47,941	6,359,327	86,057,230	1.338	49,853	6,620,059	89,542,547	1.394
7	49,125	6,406,886	83,646,750	1.344	51,040	6,675,975	87,076,266	1.405
8	44,485	6,416,651	96,677,020	1.345	46,264	6,660,483	100,563,436	1.398
9	48,765	6,437,739	87,524,940	1.360	50,276	6,630,227	90,325,738	1.393
10	47,989	6,827,568	87,044,198	1.360	The solution is simulated in iteration 3.			

* Normalized weighted sum of objectives.

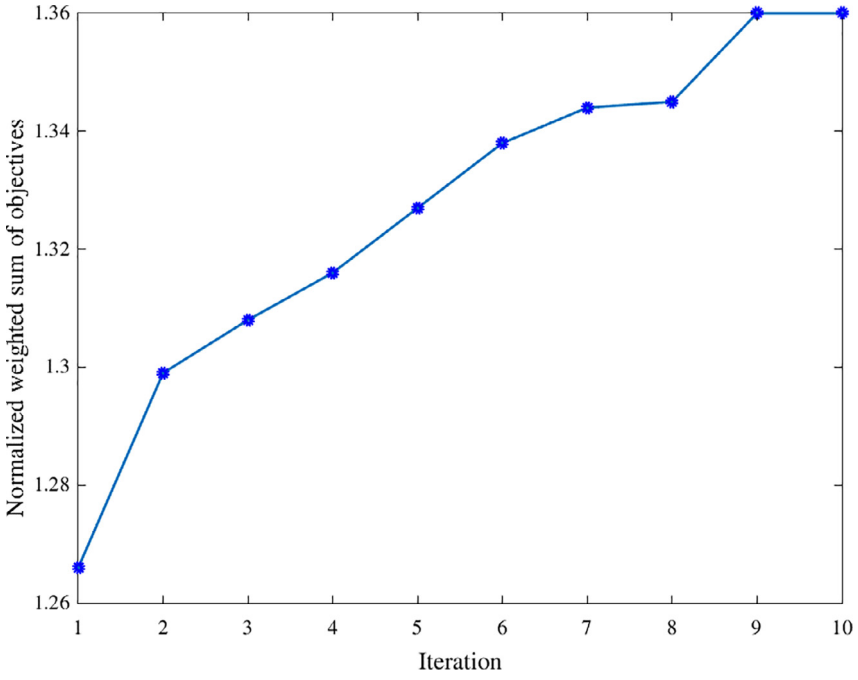


Fig. 3. Results of solving the deterministic model for problem instance 1.

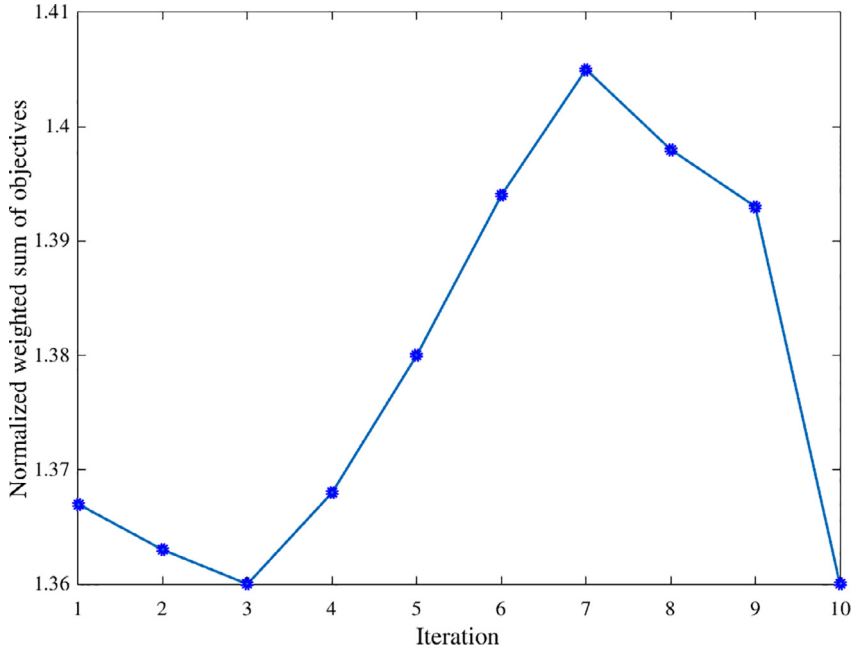


Fig. 4. Results of simulation for problem instance 1.

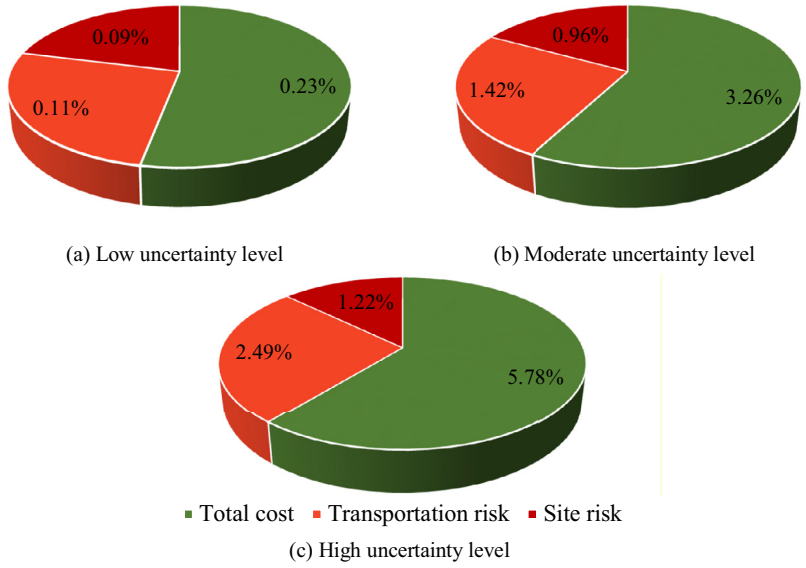


Fig. 5. Uncertainty impact under different uncertainty scenarios.

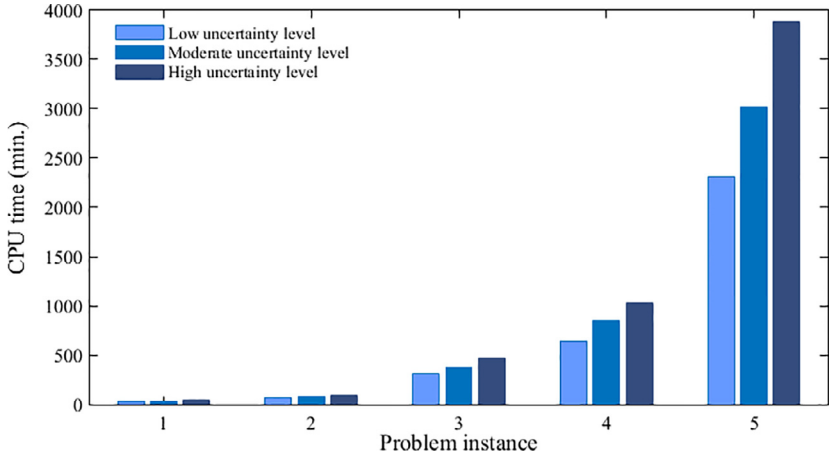


Fig. 6. Impact of uncertainty level on required CPU time in hybrid simulation-analytical modeling approach.

scenarios. It is important to note that the weight of the relative importance of each objective is taken into account when calculating the uncertainty impact.

From the results presented here, it is obvious that as the uncertainty in the stochastic parameters increases, the cost saving also increases and it is more beneficial to incorporate its impact into decision making. The reason is found in the fact that the rental collection vehicles are used when necessary to cover route failures. In better words, as the larger variances of demands are considered in route planning, the robustness of solutions will also tend to increase in order to avoid unnecessary route failures and the application of rental vehicles. Furthermore, our findings verify the conflicting nature of cost and risk concerns since gaining access to cost saving is not possible without imposing additional risks. However, reaching a balance between economic and risk issues in which the increase of risks is less than the decrease of cost is the common feature of all charts in Fig. 5.

One of the consequences of increasing the length of the uncertainty interval is the increase of the problem solving time. This is particularly evident in case of the hybrid simulation-analytical modeling approach, where an increase in the standard deviations of the parameters will increase the iteration number of the solution approach. Fig. 6 illustrates these statements. Therefore, there is a dependency relation between the problem solving time and the uncertainty level of stochastic parameters. In our proposed simheuristic approach, such changes in the uncertainty level are reflected in terms of changes in estimated standard deviation of the output parameter in Eq. (36), and subsequently in the number of scenarios.

To compare the computational performance of the solution methods, we employed two evaluation metrics, including the required CPU time and the relative percent difference between the values of the normalized weighted sum of objectives. In Fig. 7, we graphically depict the computational times required for the results presented for both solution approaches as well as the deterministic model only under moderate uncertainty level.

Our findings indicate that as the waste management network becomes larger in size, the hybrid simulation-analytical modeling approach becomes computationally very time-consuming even for small-size problems. For example, in case of problem instance 5, the algorithm can reach a stable solution in more than two-day running time. Therefore, the proposed multi-objective simheuristic approach can offer a substantial time saving (up to several hundred times) compared to another approach in solving the combinatorial

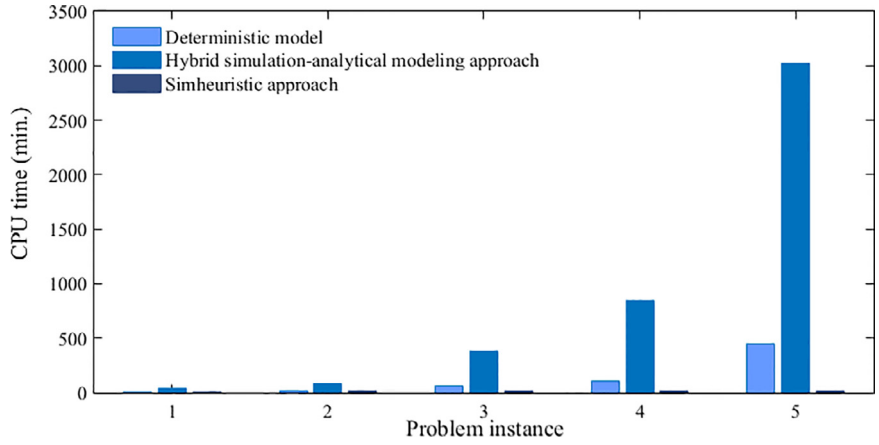


Fig. 7. Required CPU time for solving the small-scale problems with moderate uncertainty level.

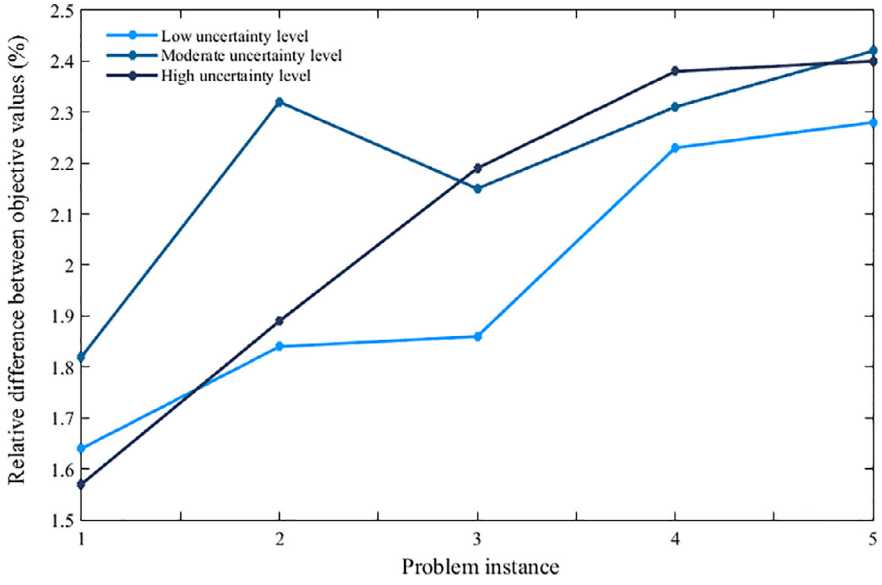


Fig. 8. Relative difference percent between objective values in small-size problems.

Table 5
Objective function values of the small-size problems under medium uncertainty level.

Problem no.	Hybrid simulation-analytical modeling approach			Proposed simheuristic approach		
	Obj. 1	Obj. 2	Obj. 3	Obj. 1	Obj. 2	Obj. 3
1	46,135	6,539,817	83,431,610	46,953	6,434,891	86,889,112
2	65,414	7,690,513	78,744,260	67,376	7,613,607	78,736,385
3	76,781	7,613,608	94,493,112	82,557	7,994,288	83,351,429
4	78,855	8,862,337	88,572,044	78,614	9,327,459	88,411,167
5	80,732	11,407,528	93,461,016	81,531	11,867,251	95,330,236

optimization problems. In case of the second evaluation metric, we calculate the relative percent difference between the values of the normalized weighted sum of objectives under different uncertainty levels to quantitatively compare the quality of the solutions obtained from both methods. The results are summarized in Fig. 8.

The diagram in Fig. 8 states the proposed method can get a solution with a relative deviation less than 2.5% from the hybrid simulation-analytical modeling approach. The precision in Fig. 8 shows that there is no meaningful relationship between the uncertainty level and the relative deviation of both methods results. However, by increasing the size of the problem space, it is observed that the relative deviation will be getting larger. The numerical results in terms of objective function values in Table 5 confirm this observation since except problem instance 5, the solutions of both approaches are non-dominated to each other.

4.3. Medium and large-size problem instances

Given the computational limitations of the commercial optimization software package in solving the combinatorial optimization problems, a limited number of problem instances are solvable using the hybrid simulation-analytical modeling approach. In this regard, and considering the proven efficiency of the proposed method in handling the small-scale problems, we will solve more problems with a larger size in order to be able to estimate the required computational time for real-size problems. Fig. 9 graphically

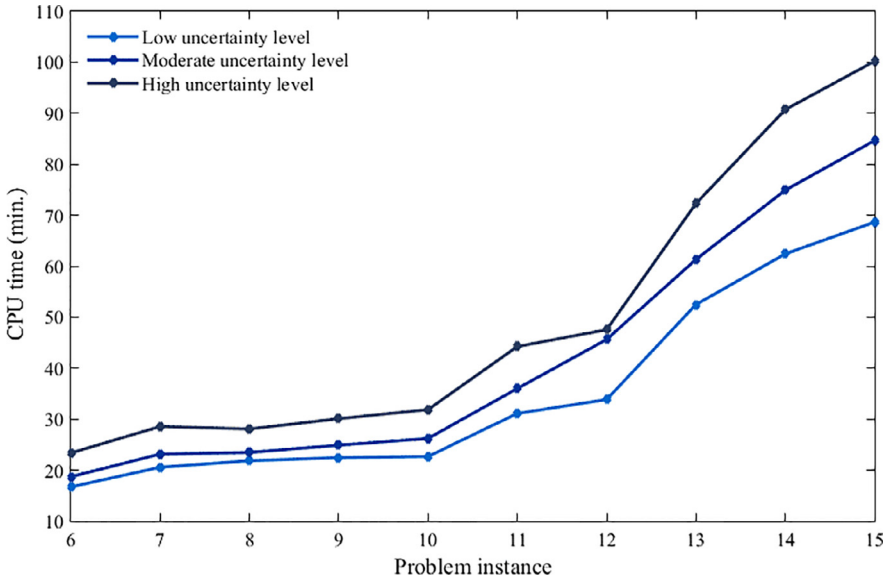


Fig. 9. Required CPU time for medium and large-size problems.

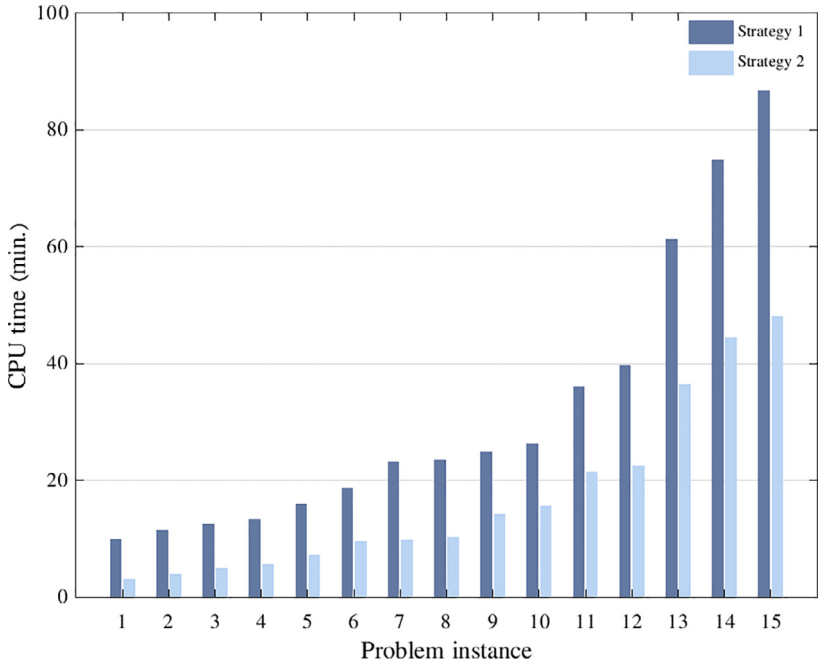


Fig. 10. Required CPU time under different solution comparison strategies.

demonstrates the upward trend in CPU time requirements when the problem size increases.

At first glance, it may seem the performance of the proposed approach in terms of computational time is unacceptable. However, the comparison of computational times of our proposed approach versus the exact method shows its relatively desirable performance in solving the real-size combinatorial optimization problems under uncertainty.

The main reasons for high time complexity of the proposed approach are associated with the number of scenarios as well as the statistical tests used for non-dominated sorting of the solutions. To investigate the effect of performing statistical tests on the time complexity, we apply the algorithm with a new structure, in which instead of comparisons using the paired t-test, the mean value of each objective under different scenarios is applied. Fig. 10 shows the solving times for the problem instances under different comparison strategies.

Fig. 10 further reveals that the statistical tests account for a significant part of the computational time in the initial configuration of the proposed approach. However, the strongest motive for applying the statistical tests in implementing the proposed approach is to achieve solutions with superior characteristics. Here, we focus only on two features of the Pareto optimal solutions, quality and spacing metrics. These metrics are explained in the following.

- Quality metric: This metric measures the proportion of Pareto solutions associated with each comparison strategy when all the obtained solutions are used to form a final Pareto solution set. Fig. 11 confirms the superiority of utilizing the statistical tests to the other approach.

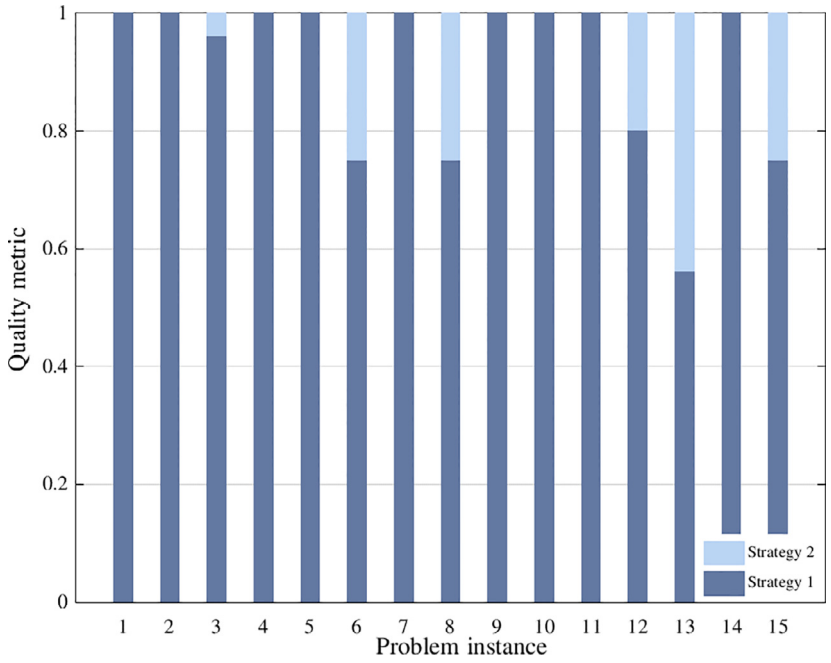


Fig. 11. Quality metric under different solution comparison strategies.

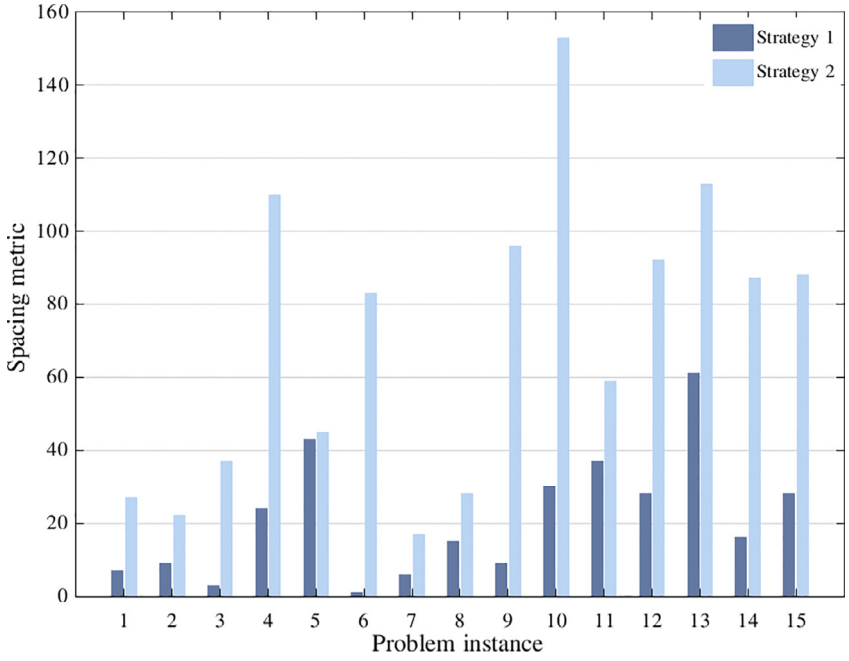


Fig. 12. Spacing metric under different solution comparison strategies.

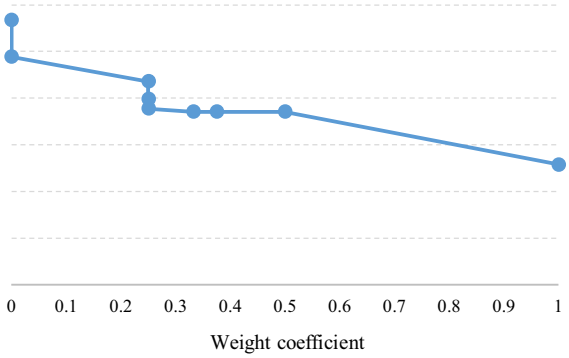
- Spacing metric: Non-uniformity of solutions as an undesirable criterion is measured by using this metric calculated on the basis of Eqs. (37) and (38). In these equations, N is the number of non-dominated solutions, $\bar{d}$ is the mean of d_i for all non-dominated solutions. The algorithm which obtains fewer values for this metric is favorable. According to Fig. 12, using statistical tests for comparing solutions guarantees the uniformity of the solutions.

$$Spacing = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(d_i - \bar{d}\right)^2} \tag{37}$$

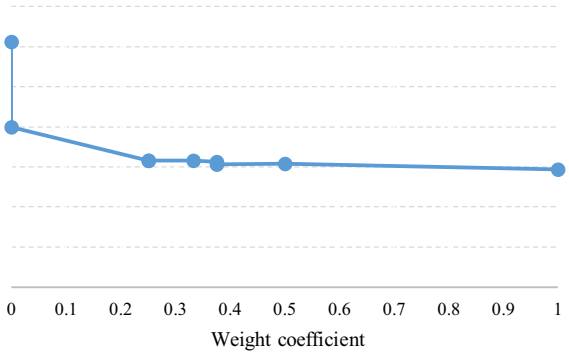
$$d_i = \min_j \sum_{t=1}^{T} \left|f_t^j - f_t^i\right|; \forall i, j \in \{1, 2, ..., N\} \tag{38}$$

4.4. Sensitivity analysis

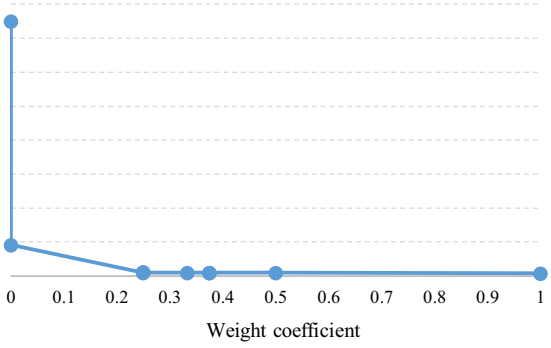
Since the implementation of both solution methods, especially the hybrid simulation-analytical modeling approach, requires setting the appropriate weight coefficients of objective functions, the purpose in this section is first to determine the best weight combination. To this end, we first define different weight combinations and solve the deterministic problem to minimize the normalized weighted sum of objectives under each weight combination scenario. Fig. 13 shows how the value of objective functions is affected



(a) Total cost function



(b) Transportation risk function



(c) Site risk function

Fig. 13. Trend of objective functions under different weight coefficient combination scenarios.

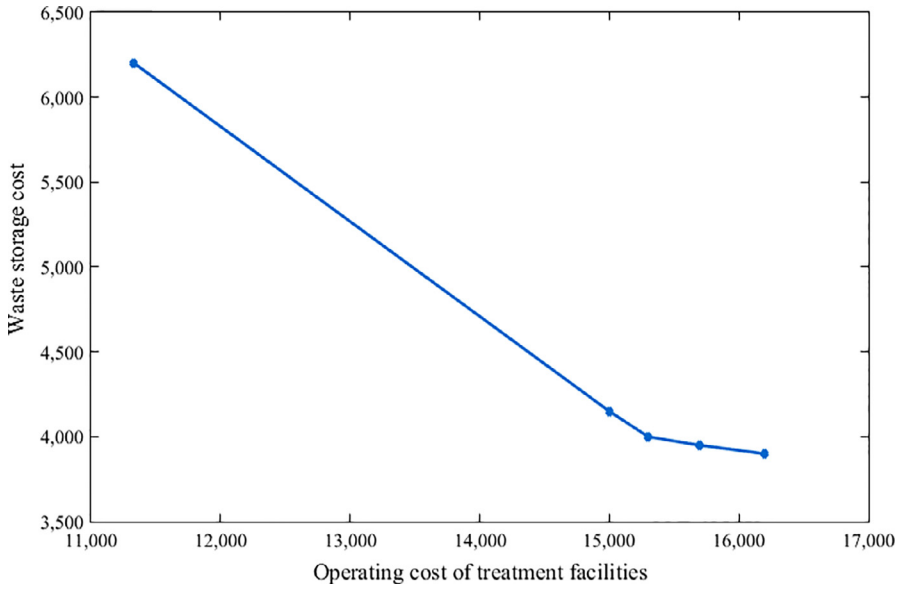


Fig. 14. Conflicting nature of operating and waste storage costs in treatment facilities.

by different weight scenarios. This analysis is done based on problem instance 1 and the results are generalized to other problem instances.

An interesting observation from Fig. 13 is that the largest changes in the objective functions occur at low weight coefficients, between 0 and 0.25. Accordingly, it seems reasonable to select the weight coefficients in such a way that the weight of all objectives is placed outside this range. In addition, the first objective function shows more sensitivity to the weight coefficients greater than 0.25 than the second and third functions. So, we set the weight vector to (0.5, 0.25, 0.25) and use it in all calculations.

The relationship between the cost of waste storage and the operating cost in treatment facilities can be illustrated as it is shown in Fig. 14. It is necessary to note that the diagram in Fig. 14 is

based on the results of previous analysis, where considering different weights for cost function leads to implementation of different management plans in treatment facilities.

As evident from Fig. 14, the greater the amount of waste storage, the lower the number of setup times, and consequently, the lower the operating cost and vice versa. Therefore, the optimal decision is to perform waste processing operations in those time periods in which the total cost of facilities has to be minimized.

5. Conclusions

As opposed to our previous work on industrial hazardous waste location-routing problem (Rabbani et al., 2018), the proposed model in this study is capable of covering inventory decisions in treatment facilities. Furthermore, incorporating uncertainty into the model through considering some parameters with random behavior and decision-making in a multi-period planning horizon are other differences that bring it closer to reality but make it more complicated. Therefore, apart from the novelty in terms of applications, the strength of the present study seems to be in terms of methodological innovations, since on one hand, the model is based on the minimization of three objectives related to the total cost, the transportation risk, and the site risk, and on the other hand, the values of the critical input data are tainted with substantial uncertainty and neglecting this uncertainty can drastically affect the overall performance of the system. With respect to this fact that the design of such systems raises hard combinatorial optimization problems, we first developed a multi-objective simheuristic approach that combines NSGA-II with Monte Carlo simulation. Then, the performance and accuracy of the solution approach were evaluated by comparing it with another recently simulation-optimization approach on a set of randomly generated test problems. The results indicated that while the second solution approach is very time-consuming, even for the small-scale problems, the proposed approach can achieve high-quality solutions in reasonable computational times. Furthermore, our experiment results clearly showed the importance of incorporating uncertainty in decision making, where a 5.78% improvement in costs can have a substantial effect on long-term performance. One interesting insight derived from the evaluation of three different uncertainty levels) ranging from 5 to 25 percent) is that the amount of improvement increases as uncertainty increases. Another consequence of increasing the uncertainty level is that the required CPU time increases with the number of scenarios and iterations in the first and second solution method, respectively. In addition, it was argued that if instead of applying the statistical tests in comparing each pair of solutions, the mean values of objectives are applied, this can lead to substantial savings in the computational time of the proposed approach. As a result, although the computational time is improved, some features of the Pareto optimal set such as quality and spacing metrics may be degraded. Finally, a sensitivity analysis was conducted and lead to identification of weight vector (0.5, 0.25, 0.25) which may have a significant influence on the outputs of the solution approaches.

With contradictory incentives, the government authorities must be cognized of the decisions of the network users and the carrier companies, and incorporate their reactions in the strategic decision-making process. This leads us to bi-level programming, which is a chance to overcome the dominant power of the regulator in the network decisions. Regarding this, in the future, we will extend this work by developing a bi-level model to accurately formulate this conflicting design process where the government and the carriers take the leader and follower positions, respectively. Also, future research could explore the application of decomposition approaches (for example, Benders decomposition algorithm) in the framework of the hybrid simulation-analytical modeling approach. In addition, considering constraints or problem characteristics associated with the hazardous material emergency response team is also another interesting direction for future research to have a resilient network.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2018.07.024.

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